# Question 1: Colored Blocks

### (a) Two Colors: Blue and Red

**Two-pointer approach:**

```
left = 0, right = n-1
while left < right:
    while A[left] == Blue: left++
    while A[right] == Red: right--
    if left < right: swap(A[left], A[right])
```

- **In-place:** Yes
- **Time:** O(n), **Space:** O(1)

### (b) Three Colors: Blue, Red, Green

```
low = 0, mid = 0, high = n-1
while mid <= high:
    if A[mid] == Blue:  swap(A[low], A[mid]); low++; mid++
    elif A[mid] == Green: mid++
    else (Red):         swap(A[mid], A[high]); high--
```

- **In-place:** Yes
- **Time:** O(n), **Space:** O(1)

### (c) Four Colors: Blue, Red, Green, Yellow

**Two passes:**

- Pass 1: Separate Blue from the rest using two-pointer  O(n)
- Pass 2: Apply Dutch National Flag on the remaining {Red, Green, Yellow} portion  O(n)
- **In-place:** Yes
- **Time:** O(n), **Space:** O(1)

## Question 2: QuickSort (Median-of-Three)

**(a) {1, 2, 3, 4, 5, 6, 7, 8, 9}**

- Pivot = median(1, 5, 9) = **5**  {1,2,3,4} | **5** | {6,7,8,9}
- Left: pivot = median(1,3,4) = **3** {1,2} | **3** | {4}  {1} | **2**
- Right: pivot = median(6,8,9) = **8**  {6,7} | **8** | {9}  {6} | **7**

**Result: {1, 2, 3, 4, 5, 6, 7, 8, 9}**

**(b) {8, 7, 6, 5, 4, 3, 2, 1, 9}**

- Pivot = median(8, 4, 9) = **8**  {7,6,5,4,3,2,1} | **8** | {9}
- Left: pivot = median(7,4,1) = **4** {3,2,1} | **4** | {7,6,5}
    - {3,2,1}: pivot=**2** = {1} | **2** | {3}
    - {7,6,5}: pivot=**6**  {5} | **6** | {7}

**Result: {1, 2, 3, 4, 5, 6, 7, 8, 9}**

**(c) {9, 1, 8, 2, 7, 3, 6, 4, 5}**

- Pivot = median(9, 7, 5) = **7** {1,2,3,6,4,5} | **7** | {9,8}
- Left: pivot = median(1,3,5) = **3** {1,2} | **3** | {6,4,5}
    - {1,2}: pivot=**1**  **1** | {2}
    - {6,4,5}: pivot=**5**  {4} | **5** | {6}
- Right: {9,8}  pivot=**8**  **8** | {9}

**Result: {1, 2, 3, 4, 5, 6, 7, 8, 9}**

## Question 3: QuickSelect (Median-of-Three)

**(a) {1, 2, 3, 4, 5, 6, 7, 8, 9}, k = 4**

- Pivot = median(1,5,9) = **5**, lands at index 4. k=4  recurse left on {1,2,3,4}
- Pivot = median(1,3,4) = **3**, lands at index 2. k=4  recurse right on {4}

- Single element → **Answer = 4**

**(b) {8, 7, 6, 5, 4, 3, 2, 1, 9}, k = 5**

- Pivot = median(8,4,9) = **8**, lands at index 7. k=5  recurse left on {7,6,5,4,3,2,1}
- Pivot = median(7,4,1) = **4**, lands at index 3. k=5  recurse right on {7,6,5}
- Pivot = median(7,6,5) = **6**, lands at index 1. k=5  recurse left on {5}
- Single element  **Answer = 5**

**(c) {9, 1, 8, 2, 7, 3, 6, 4, 5}, k = 6**

- Pivot = median(9,7,5) = **7**, lands at index 6. k=6  recurse left on {1,2,3,6,4,5}
- Pivot = median(1,3,5) = **3**, lands at index 2. k=6  recurse right on {6,4,5}
- Pivot = median(6,4,5) = **5**, lands at index 1. k=6  recurse right on {6}
- Single element → **Answer = 6**